

EE 445/645: Physical Models in Remote Sensing (Spring 2026)

Chapter 04-Part 02: Method of Successive Orders of Scattering – Notes (See C4-P2 PPTs)

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Slide 1 — Class Overview

Topics Covered in This Lecture

This lecture, Part 02 of Chapter 04, develops the Successive Orders of Scattering (SOS) approximation — the primary method used to solve the collided radiative transfer problem in vegetation canopies. Building on the uncollided field solved analytically in Part 01, we now address the far harder question of what happens after photons begin interacting with canopy elements.

The lecture is organised as follows:

- Separation into Uncollided and Collided Problems
- Solution of the Uncollided Problem (review)
- The Collided Problem — its governing equation and why it cannot be solved analytically
- Method of Successive Orders of Scattering — the key idea
- First-Collision Problem — $I^1(L, \Omega)$ and its analytical solution
- Graphical description of the integration along a ray
- Example results for the first-collision approximation
- Higher-Order Collision Problems — $I^2(L, \Omega)$, $I^3(L, \Omega)$, ..., $I^n(L, \Omega)$
- Example results for the full SOS solution
- Concluding remarks — strengths and weaknesses of the method

By the end of this lecture you should be able to derive the SOS decomposition from first principles, write the boundary-value problem for each scattering order, evaluate the path-integral solution geometrically, and interpret the energy balance tables produced by the SOS code.

Decomposing the Total Intensity Field

The starting point is a decomposition of the total specific intensity $I(L, \Omega)$ at canopy depth L in direction Ω into two physically distinct components:

$$I(L, \Omega) = I^0(L, \Omega) + I^c(L, \Omega)$$

where:

- $I^0(L, \Omega)$ is the uncollided intensity — photons that have travelled from the boundary (canopy top or soil) to depth L along direction Ω without interacting with any leaf. These photons follow pure Beer-Lambert attenuation and their intensity is known analytically from Part 01.
- $I^c(L, \Omega)$ is the collided intensity — photons that have experienced one or more collisions (scattering or absorption events) with canopy elements before reaching (L, Ω) . This is the unknown we need to solve for.

This decomposition is exact — no approximation has been made yet. It simply recognises that the two populations of photons satisfy different equations:

- The uncollided problem can be solved analytically because the source term is zero: photons in I^0 travel in straight lines, attenuating exponentially. The solution is available at every depth L and every direction Ω .
- The collided problem cannot be solved analytically because the scattered intensity I^c appears on both the left-hand side (as the unknown) and on the right-hand side (inside the scattering integral as a source). This is a coupled integro-differential equation.

The analytical solution for I^0 from Part 01 is therefore an essential prerequisite — it provides the "driving term" for the collided problem. Once I^0 is known everywhere, the first-collision source $Q(L, \Omega)$ can be evaluated, and the SOS algorithm can begin.

Uncollided Radiative Transfer: RED Band ($\omega L = 0.10$)

This slide presents Figure 04 of the course report — the uncollided radiation field for the RED spectral band. The canopy and illumination parameters are:

- $f_{dir} = 0.7$ (70% direct solar beam, 30% isotropic diffuse sky)
- $\theta_o = 140^\circ$, $\mu_o = -0.766$ (sun below the horizon from above — downward travelling)
- $\phi_o = 0^\circ$, $G = 0.5$ (spherical leaf distribution), $F_{in} = 1.0 \text{ W m}^{-2}$
- $\rho_L = 0.06$, $\tau_L = 0.04$, $\omega L = \rho_L + \tau_L = 0.10$ (low single-scattering albedo — mostly absorbing)
- $p_g = 0.10$ (soil hemispherical reflectance)
- $LAI = 1.5$ (blue) and $LAI = 4.0$ (red)

Three panels are shown:

- Plot A — Downward uncollided flux $F_d^o(L)/F_{in}$ vs normalised depth L/LAI . Both direct (solid) and diffuse (dashed) components are plotted. The direct beam decays exponentially from F_{in} at $L=0$ to about 0.09 ($LAI=1.5$) and 0.01 ($LAI=4.0$) at the base. The diffuse component is lower and decreases more slowly because photons arrive from all downward angles, and near-horizontal paths are attenuated less rapidly.
- Plot B — Upward uncollided flux $F_u^o(L)/F_{in}$ vs L/LAI . This is the ground-reflected component propagating back up through the canopy. It is small ($p_g = 0.10$ and $\omega L = 0.10$ so little light reaches the ground and little is reflected). F_u^o increases from base to top as the attenuating canopy path shortens.
- Plot C — Uncollided BRF at $L=0$ in the principal plane ($\phi=0^\circ$ forward scatter, $\phi=180^\circ$ backscatter). The BRF is very small (peak ~ 0.016) because nearly all incident radiation is absorbed on the first interaction. The hotspot at $\theta = 40^\circ$ (backscatter direction, same as the solar zenith angle projected onto the principal plane) is clearly visible.

Figure 04 of the course report. Code: C4-P1-Uncollided-RT.py

Uncollided Radiative Transfer: NIR Band ($\omega_L = 0.975$)

This slide presents Figure 05 of the report — the same uncollided calculation for the NIR band. The leaf optical properties change dramatically:

- $\rho_L = 0.525$, $\tau_L = 0.45$, $\omega_L = 0.975$ (nearly conservative scatterer — only 2.5% absorbed per interaction)
- $p_g = 0.20$ (brighter soil in NIR)

Key observations compared to the RED band:

- Plot A (downward flux): nearly identical to RED for the uncollided component. This is expected — the Beer-Lambert attenuation of I^0 depends on $G(\Omega)$ and LAI, not on the leaf single-scattering albedo ω_L . I^0 is the same regardless of whether leaves absorb or scatter.
- Plot B (upward flux): again very similar to RED for the uncollided component, for the same reason. However, the absolute values of the ground-reflected flux are slightly higher because $p_g = 0.20 > 0.10$.
- Plot C (BRF): the uncollided BRF is slightly higher (peak ~ 0.035 vs 0.016) because more light reaches the ground and more is reflected back with $p_g = 0.20$. The hotspot geometry is identical.

The critical insight is this: the uncollided field is essentially the same for RED and NIR. The dramatic difference between RED and NIR results — large reflectance, high transmittance, many scattering orders needed — will emerge entirely from the collided problem, driven by ω_L . The SOS algorithm on the following slides makes this explicit.

Figure 05 of the course report. Code: C4-P1-Uncollided-RT.py

Governing Equation for the Collided Intensity

Substituting the decomposition $I = I^0 + I^c$ into the full 1D RTE and using the fact that I^0 satisfies the uncollided equation, we obtain the boundary-value problem for $I^c(L, \Omega)$:

$$-\mu \partial/\partial L I^c(L, \Omega) + G(\Omega) I^c(L, \Omega) = Q(L, \Omega) + S(L, \Omega)$$

Upper boundary condition (no incoming collided radiation from above):

$$I^c(L=0, \Omega) = 0, \quad \mu < 0$$

Lower boundary condition (Lambertian soil reflects the collided downward flux):

$$I^c(L=LH, \Omega) = (1/\pi) \int_2 \pi^- d\Omega' \rho_s(\Omega' \rightarrow \Omega) |\mu'| I^c(L=LH, \Omega'), \quad \mu > 0$$

The two source terms on the right-hand side are physically distinct:

- $Q(L, \Omega)$ — the first-collision source. This is the rate at which previously uncollided photons I^0 scatter for the first time into direction Ω at depth L . Because I^0 is already known analytically, Q is a known function:

$$Q(L, \Omega) \equiv (1/\pi) \int_4 \pi d\Omega' \Gamma(\Omega' \rightarrow \Omega) I^0(L, \Omega')$$

where $\Gamma(\Omega' \rightarrow \Omega)$ is the area scattering phase function (bi-Lambertian for spherical leaves).

- $S(L, \Omega)$ — the distributed (multiple-collision) source. This is the rate at which already-collided photons I^c scatter again into direction Ω :

$$S(L, \Omega) = (1/\pi) \int_4 \pi d\Omega' \Gamma(\Omega' \rightarrow \Omega) I^c(L, \Omega')$$

The fundamental difficulty is that S contains I^c — the very quantity we are trying to solve for. The collided problem is therefore an integro-differential equation that cannot be solved analytically. This is why we resort to the Successive Orders of Scattering approximation.

The Key Idea: Splitting by Scattering Order

The SOS method resolves the difficulty of the coupled collided problem by expanding I^c as a sum of contributions from photons that have scattered exactly n times:

$$I^c(L, \Omega) = I^1(L, \Omega) + I^2(L, \Omega) + I^3(L, \Omega) + \dots + I^n(L, \Omega)$$

where:

- $I^1(L, \Omega)$ — intensity of once-scattered photons (first collision)
- $I^2(L, \Omega)$ — intensity of twice-scattered photons (second collision)
- $I^n(L, \Omega)$ — intensity of photons that have scattered exactly n times

This expansion is exact in the limit $n \rightarrow \infty$. Inserting it into the collided RTE and matching terms order by order produces a chain of n decoupled boundary-value problems, each of the same form as the original RTE but with a known source:

- The source for I^1 is $Q(L, \Omega)$ — which is known because I^0 is known.
- The source for I^2 is the scattering of I^1 — which is known once I^1 is solved.
- The source for I^n is the scattering of I^{n-1} — known once I^{n-1} is solved.

Each problem has zero boundary condition at the top (no n th-order photons entering from outside) and a Lambertian reflection condition at the soil. Crucially, each problem has an analytical (path-integral) solution — the same structure as the uncollided problem, but driven by the previous-order intensity instead of an external source.

The algorithm is therefore sequential: solve I^1 , then use it to form the source for I^2 , solve I^2 , use it for I^3 , and so on until the cumulative sum I^c converges. Convergence is rapid when ωL is small (each order is attenuated by a factor $\approx \omega L \cdot G \approx \omega L/2$), but slow when $\omega L \approx 1$ (NIR), where many orders may be needed.

Boundary-Value Problem for Once-Scattered Photons

Inserting I^1 into the collided RTE and noting that $S = 0$ (once-scattered photons have not yet scattered a second time), the $I^1(L, \Omega)$ boundary-value problem is:

$$-\mu \partial/\partial L I^1(L, \Omega) + G(\Omega) I^1(L, \Omega) = (1/\pi) \int_4 \pi d\Omega' \Gamma(\Omega' \rightarrow \Omega) I^0(L, \Omega')$$

with boundary conditions:

$$I^1(L=0, \Omega) = 0, \quad \mu < 0$$

$$I^1(L=LH, \Omega) = (1/\pi) \int_2 \pi^- d\Omega' \rho_S(\Omega' \rightarrow \Omega) |\mu'| I^1(L=LH, \Omega'), \quad \mu > 0$$

The right-hand side is the first-collision source term $Q(L, \Omega)$:

$$Q(L, \Omega) \equiv (1/\pi) \int_4 \pi d\Omega' \Gamma(\Omega' \rightarrow \Omega) I^0(L, \Omega')$$

This source is fully determined because $I^0(L, \Omega')$ is known at all (L, Ω') . In practice it has two contributions, evaluated separately:

- Direct solar beam: $Q_{\text{dir}}(L, \Omega) = (1/\pi) \Gamma(\Omega_0 \rightarrow \Omega) \cdot I_0 \cdot \exp(-G \cdot L / |\mu_0|)$. This is the single-scattering of the attenuated direct beam into direction Ω .
- Diffuse sky: $Q_{\text{dif}}(L, \Omega) = (1/\pi) \sum_i \sum_j \Gamma(\Omega_{ij} \rightarrow \Omega) \cdot I_d \cdot \exp(-G \cdot L / |\mu_i|) \cdot w_i \cdot w_j$, evaluated by numerical quadrature over all downward directions.

Once Q is computed, the PDE for I^1 has an analytical path-integral solution — the photon transport equation integrated along the ray direction. This is the most important structural result of the first-collision problem.

Path-Integral Solution for Once-Scattered Photons

The governing PDE for $I'(L, \Omega)$ is a first-order linear ODE along the ray direction. Its exact solution is a path integral of the source $Q(L', \Omega)$ along the ray, weighted by the probability of the photon surviving (without further collision) from L' to the observation point L .

For downward directions ($\mu < 0$), integrating from the canopy top ($L'=0$) down to L :

$$I^1(L, \Omega) = (1/|\mu|) \int_0^L dL' Q(L', \Omega) \cdot P[\Omega, (L-L')], \quad \mu < 0$$

For upward directions ($\mu > 0$), integrating from the soil ($L'=LH$) up to L , including the ground reflection term:

$$I^1(L, \Omega) = (1/|\mu|) \int^{LH} dL' Q(L', \Omega) \cdot P[\Omega, (L'-L)] + I^1(LH, \Omega) \cdot P[\Omega, (LH-L)], \quad \mu > 0$$

The function $P[\Omega, (L_2-L_1)]$ is the single-path transmittance — the probability that a photon travelling in direction Ω is not intercepted by any leaf over the canopy path length $(L_2-L_1)/|\mu|$:

$$P[\Omega, (L_2-L_1)] = \exp[-(1/|\mu|) G(\Omega) (L_2-L_1)]$$

For the spherical leaf distribution, $G(\Omega) = 0.5$ for all directions, giving:

$$P[\Omega, (L_2-L_1)] = \exp[-0.5 (L_2-L_1) / |\mu|]$$

Physical interpretation: the once-scattered intensity at (L, Ω) is the sum over all points L' above (or below) L of the source $Q(L', \Omega)$ — the rate at which photons at L' first scatter into Ω — multiplied by the survival probability P from L' to L . The $1/|\mu|$ factor converts the depth integral to a path length.

For weakly scattering media ($\omega L < 0.1$, as in the RED band), the single-scattering approximation — retaining only I^1 — may be sufficient to describe the scene with acceptable accuracy.

Physical Picture: Accumulation of Scattered Photons Along a Downward Ray

The path-integral solution for downward directions has a clear geometric interpretation, illustrated by the diagram on the slide.

Consider evaluating $I^1(L, \Omega)$ for a downward direction Ω ($\mu < 0$). At each intermediate depth L' between 0 and L , first-collision photons are being created by the source $Q(L', \Omega)$ — uncollided photons I^0 arriving from all directions and scattering for the first time into Ω . These newly scattered photons then travel downward along Ω from L' to L over a path length $(L-L')/|\mu|$, surviving with probability $P[\Omega, (L-L')]$.

The integral accumulates these contributions from all depths L' from 0 to L :

$$I^1(L, \Omega) = (1/|\mu|) \int_0^L dL' Q(L', \Omega) \cdot \exp[-G(\Omega)(L-L')/|\mu|], \quad \mu < 0$$

On the diagram:

- The vertical axis represents canopy depth from $L=0$ (top) to $L=LH$ (bottom).
- A red circle at depth L' marks a scattering event — a photon from I^0 scatters into direction Ω for the first time.
- The blue arrow shows the subsequent propagation of that once-scattered photon from L' down to L along direction Ω .
- The dashed segment $(L-L')$ labels the remaining propagation distance.
- At L , the accumulated once-scattered flux $I^1(L, \Omega)$ is the sum of all such contributions.

At the bottom of the diagram the two key formulas are shown together:

- $P[\Omega, (L_2-L_1)] = \exp[-(1/|\mu|) \cdot G(\Omega) \cdot (L_2-L_1)]$ — the path transmittance.
- $Q(L, \Omega) = (1/\pi) \int_4\pi d\Omega' \Gamma(\Omega' \rightarrow \Omega) I^0(L, \Omega')$ — the first-collision source.

Physical Picture: Accumulation of Scattered Photons Along an Upward Ray

For upward directions ($\mu > 0$) the geometry is mirrored: integration runs from the soil surface upward to the observation depth L .

$$I^1(L, \Omega) = (1/|\mu|) \int^{LH} dL' Q(L', \Omega) \cdot P[\Omega, (L'-L)] + I^1(LH, \Omega) \cdot P[\Omega, (LH-L)], \quad \mu > 0$$

The two terms have distinct physical meanings:

- Volume term — $(1/|\mu|) \int^{LH} dL' Q(L', \Omega) \cdot P[\Omega, (L'-L)]$: scattered photons created at every depth L' between L and LH , propagating upward to L . The path from L' to L has length $(L'-L)/\mu$, and survival probability $P[\Omega, (L'-L)]$.
- Surface term — $I^1(LH, \Omega) \cdot P[\Omega, (LH-L)]$: photons of once-scattered intensity that strike the soil at $L=LH$ and are reflected back upward by the Lambertian soil in direction Ω , then propagate from LH to L , surviving with probability $P[\Omega, (LH-L)]$.

On the diagram:

- The upward-pointing blue arrow shows a once-scattered photon travelling from a scattering event at L (red circle) up to the observation point — in the upward case, the source is at depth $L' > L$ and the photon travels toward the top.
- The red dot at LH represents the ground reflection contributing the surface term.

Both downward and upward solutions share the same mathematical structure: a path integral of Q weighted by survival probability P , plus a boundary term (zero at the top, ground reflection at the bottom). This structure also holds for all higher scattering orders $I^2, I^3, \dots, I^n$ — only the source changes.

Energy Balance: RED Band, $I^0 + I^1$ (First-Collision Approximation)

This table shows the energy balance for the RED band when only the uncollided (I^0) and first-collision (I^1) intensities are included. Parameters: $\omega L = 0.100$, $\rho_g = 0.1$, $\rho_L = 0.06$, $\tau_L = 0.04$.

Convergence is enforced on BOTH boundary intensities AND scalar irradiance (max relative change < 1%). Absorption is computed from the scalar irradiance: $Abs = G \cdot (1 - \omega L) \cdot \int S(L) dL$, which is independent of Reflected and Transmitted.

Results for LAI = 1.5:

- Reflected: 0.0081 (Direct) + 0.0029 (Diffuse) + 0.9890 (Collided) = 0.0305 total
- Transmitted: 0.2367 + 0.0836 + 0.0104 = 0.3307 total
- Absorbed: 0.6332 (from scalar irradiance)
- SUM: 0.9944 → Imbalance = -0.562%

Results for LAI = 4.0:

- Reflected: 0.0003 + 0.0001 + 0.9996 = 0.0217 total
- Transmitted: 0.0463 + 0.0163 + 0.0044 = 0.0670 total
- Absorbed: 0.9044 (from scalar irradiance)
- SUM: 0.9930 → Imbalance = -0.696%

Interpretation: the imbalance (-0.56% and -0.70%) represents the energy still residing in uncomputed higher-order scatterings $I^2 + I^3 + \dots$. For RED with $\omega L = 0.10$, this is very small — leaves absorb nearly all light at the first collision, so higher scattering orders contribute little. The first-collision approximation is essentially sufficient for RED.

Imbalance = energy in $I_2 + I_3 + \dots$ not yet included (first-collision).

Energy Balance: NIR Band, $I^0 + I^1$ (First-Collision Approximation)

The same first-collision calculation for the NIR band reveals a dramatically different picture. Parameters: $\omega_L = 0.975$, $\rho_g = 0.2$, $\rho_L = 0.525$, $\tau_L = 0.45$.

Results for LAI = 1.5:

- Reflected: $0.0163 + 0.0057 + 0.9780 = 0.2046$ total
- Transmitted: $0.2104 + 0.0743 + 0.0982 = 0.3829$ total
- Absorbed: 0.0260 (from scalar irradiance)
- SUM: $0.6135 \rightarrow \text{Imbalance} = -38.654\%$

Results for LAI = 4.0:

- Reflected: $0.0006 + 0.0002 + 0.9992 = 0.1952$ total
- Transmitted: $0.0411 + 0.0145 + 0.0420 = 0.0976$ total
- Absorbed: 0.0404 (from scalar irradiance)
- SUM: $0.3332 \rightarrow \text{Imbalance} = -66.677\%$

Interpretation: the imbalance is enormous — -39% and -67% respectively. This means that at LAI=4.0 only one-third of the incident energy is accounted for after just one scattering order. The remaining two-thirds is still being carried by photons that have scattered two, three, or more times but have not yet been computed.

This is the direct physical consequence of $\omega_L = 0.975$. Each collision returns 97.5% of intercepted energy back into the radiation field. A photon in an NIR canopy may bounce dozens of times before being absorbed or escaping. The first-collision approximation is completely inadequate for NIR — many scattering orders are required.

This slide is the most powerful pedagogical demonstration of why the full SOS iteration is necessary for high-albedo canopies.

Imbalance = energy in $I_2 + I_3 + \dots$ not yet included (first-collision).

Extending to Twice-Scattered Photons

With I^1 fully computed, we can form the second-collision source term $Q^2(L, \Omega)$ and solve for $I^2(L, \Omega)$. The boundary-value problem for I^2 has exactly the same structure as for I^1 , but with I^1 driving the source instead of I^0 :

$$-\mu \partial/\partial L I^2(L, \Omega) + G(\Omega) I^2(L, \Omega) = (1/\pi) \int_4 \pi d\Omega' \Gamma(\Omega' \rightarrow \Omega) I^1(L, \Omega')$$

Boundary conditions:

$$I^2(L=0, \Omega) = 0, \quad \mu < 0$$

$$I^2(L=LH, \Omega) = (1/\pi) \int_2 \pi^- d\Omega' \rho s(\Omega' \rightarrow \Omega) |\mu'| I^2(L=LH, \Omega'), \quad \mu > 0$$

The right-hand side is the second-collision source $Q^2(L, \Omega)$:

$$Q^2(L, \Omega) = (1/\pi) \int_4 \pi d\Omega' \Gamma(\Omega' \rightarrow \Omega) I^1(L, \Omega')$$

This can be evaluated because $I^1(L, \Omega')$ is now known from the previous step. The PDE for $I^2(L, \Omega)$ then has the same analytical path-integral solution as I^1 :

$$I^2(L, \Omega) = (1/|\mu|) \int_0^L dL' Q^2(L', \Omega) \cdot P[\Omega, (L-L')], \quad \mu < 0$$

and analogously for upward directions.

The cumulative collided intensity after two orders is:

$$I^c \approx I^1(L, \Omega) + I^2(L, \Omega)$$

The pattern is now clear: each scattering order I^n is sourced by the scattering of I^{n-1} , and all problems share the same analytical structure. The SOS is a sequential, iterative algorithm where each step is no more complex than the previous one.

nth-Order Scattering and the Complete Solution

The general pattern is: once $I^{n-1}(L, \Omega)$ is known, the nth-order problem is:

$$-\mu \partial/\partial L I^n(L, \Omega) + G(\Omega) I^n(L, \Omega) = (1/\pi) \int_4 \pi d\Omega' \Gamma(\Omega' \rightarrow \Omega) I^{n-1}(L, \Omega')$$

with boundary conditions:

$$I^n(L=0, \Omega) = 0, \quad \mu < 0$$

$$I^n(L=LH, \Omega) = (1/\pi) \int_2 \pi^- d\Omega' \rho_s(\Omega' \rightarrow \Omega) |\mu'| I^n(L=LH, \Omega'), \quad \mu > 0$$

The right-hand side is the nth-order collision source $Q^n(L, \Omega) = (1/\pi) \int \Gamma I^{n-1} d\Omega'$, which is known because I^{n-1} has already been computed.

The total collided intensity is then:

$$I^c(L, \Omega) = I^1(L, \Omega) + I^2(L, \Omega) + I^3(L, \Omega) + \dots + I^n(L, \Omega)$$

and the total intensity, including the uncollided component:

$$I(L, \Omega) = I^0(L, \Omega) + I^c(L, \Omega)$$

The SOS series converges because each successive order I^n is smaller than I^{n-1} by a factor approximately equal to the mean single-scattering albedo times the geometry factor:

$$||I^n|| / ||I^{n-1}|| \approx \omega L \cdot G = \omega L / 2$$

For RED ($\omega L = 0.10$): each order is ~5% of the previous. Convergence is very fast — typically 3 orders suffice.

For NIR ($\omega L = 0.975$): each order is ~49% of the previous. Convergence is slow — 8 to 16 orders may be needed depending on LAI.

In the code, convergence is determined by a dual criterion: (A) boundary intensities and (B) scalar irradiance profile must both change by less than 1% before the iteration stops.

Full SOS Solution: RED Band ($I^0 + I^1 + \dots + I^n$, $n=3$)

This slide shows the complete SOS solution for the RED band after convergence ($n=3$ orders for both LAI values). Four panels are displayed:

- Plot A — Downward flux profile $F_d(L)/F_{in}$ vs L/LAI . Solid lines show the total ($I^0 + I^1$); dotted lines show the uncollided I^0 alone. The difference is small for RED — the collided contribution is modest because $\omega L = 0.10$. The denser canopy ($LAI=4.0$, red) attenuates the flux more steeply.
- Plot B — Upward flux profile $F_u(L)/F_{in}$ vs L/LAI . Again the difference between total and uncollided is small for RED. F_u is significantly smaller than F_d at every depth, reflecting high canopy absorption.
- Plot C — BRF at $L=0$ in the principal plane. The total BRF (solid) is only marginally larger than the uncollided BRF (dotted), confirming that single scattering dominates in the RED band. The hotspot at $\theta \approx 40^\circ$ (backscatter) is the dominant feature.
- Plot D — Canopy absorption profile $dAbs/dL = -d(F_d - F_u)/dL$ vs normalised depth L/LAI (depth increases downward). The absorption rate is highest at the top of the canopy where the radiation field is most intense, and decreases monotonically with depth. For $LAI=4.0$ (red), absorption is concentrated in the upper layers — almost all light is absorbed before reaching the bottom.

The key message for RED: the first-collision approximation (slides 11) already gives essentially the correct answer. Only a small correction comes from I^2 , I^3 .

Full SOS Solution: NIR Band ($I^0 + I^1 + \dots + I_n$, $n=8$ and 16)

The full SOS solution for NIR is qualitatively different from RED in every panel. Convergence requires $n=8$ orders for $LAI=1.5$ and $n=16$ orders for $LAI=4.0$.

- Plot A — Downward flux. The total F_d is dramatically higher than the uncollided I^0 at every depth. Multiple scattering returns substantial downward flux that was lost from the direct and diffuse beams. The denser canopy ($LAI=4$) attenuates I^0 strongly but the total F_d remains significant because scattered photons keep propagating downward.
- Plot B — Upward flux. F_u is large — the canopy is highly reflective in NIR, and substantial upward flux builds up throughout the canopy depth. The total F_u can reach 40–50% of F_{in} at the top for $LAI=1.5$.
- Plot C — BRF. Dramatically larger than the uncollided BRF. The total BRF reaches 0.6–0.7 for $LAI=4.0$, compared to ~ 0.035 for the uncollided component. Multiple scattering completely dominates the NIR reflectance. The shape is smoother — the hot spot is less pronounced relative to the overall BRF level.
- Plot D — Absorption profile. The absorbed flux per unit LAI is very small ($< 0.1 \text{ W m}^{-2}$ per unit LAI , compared to > 0.6 for RED). The absorption decreases with depth — what little absorption occurs happens mostly in the upper canopy. For $LAI=4.0$, the profile is remarkably flat in the upper half, then drops sharply near the bottom.

The contrast between RED and NIR in these four panels illustrates the full dynamic range of canopy radiative behaviour as a function of single-scattering albedo.

Energy Balance: RED Band, Full SOS ($I^0 + I^1 + \dots + I_n$)

Parameters: $\omega L = 0.100$, $\rho_g = 0.1$, $\rho_L = 0.06$, $\tau_L = 0.04$. Convergence: dual criterion (boundary intensities AND scalar irradiance < 1%). $n=3$ orders for both LAI values.

Results for LAI = 1.5:

- Reflected: $0.0081 + 0.0029 + 0.9890 = 0.0314$ total
- Transmitted: $0.2367 + 0.0836 + 0.0111 = 0.3314$ total
- Absorbed: 0.6350 (scalar irradiance)
- SUM: $0.9978 \rightarrow \text{Imbalance} = -0.218\%$

Results for LAI = 4.0:

- Reflected: $0.0003 + 0.0001 + 0.9996 = 0.0228$ total
- Transmitted: $0.0463 + 0.0163 + 0.0049 = 0.0674$ total
- Absorbed: 0.9094 (scalar irradiance)
- SUM: $0.9997 \rightarrow \text{Imbalance} = -0.027\%$

Interpretation: for RED the full SOS converges to an excellent energy balance. The residual imbalance of -0.2% and -0.03% is the GL quadrature limitation on the scalar irradiance for the multiply-scattered field — it is not a convergence failure. Ref and Trans are accurate (boundary criterion also satisfied).

Comparing with the first-collision result (slide 11): Absorbed increases only marginally from 0.633 to 0.635 (LAI=1.5) and from 0.904 to 0.909 (LAI=4.0). The full SOS barely changes the RED answer — confirming that nearly all the physics is captured in the first-collision approximation for this band.

Residual imbalance reflects GL quadrature limitation on scalar irradiance for high- ωL . Ref and Trans are accurate (boundary criterion also satisfied).

Energy Balance: NIR Band, Full SOS ($I^0 + I^1 + \dots + I_n$)

Parameters: $\omega L = 0.975$, $\rho_g = 0.2$, $\rho_L = 0.525$, $\tau_L = 0.45$. Convergence: dual criterion. $n=8$ orders ($LAI=1.5$), $n=16$ orders ($LAI=4.0$).

Results for $LAI = 1.5$:

- Reflected: $0.0163 + 0.0057 + 0.9780 = 0.3912$ total
- Transmitted: $0.2104 + 0.0743 + 0.2329 = 0.5175$ total
- Absorbed: 0.0371 (scalar irradiance)
- SUM: $0.9458 \rightarrow$ Imbalance = -5.417%

Results for $LAI = 4.0$:

- Reflected: $0.0006 + 0.0002 + 0.9992 = 0.5584$ total
- Transmitted: $0.0411 + 0.0145 + 0.2589 = 0.3145$ total
- Absorbed: 0.0971 (scalar irradiance)
- SUM: $0.9701 \rightarrow$ Imbalance = -2.995%

Interpretation: comparing with the first-collision result (slide 12) the transformation is dramatic. Reflectance jumped from 0.20 to 0.39 ($LAI=1.5$) and from 0.20 to 0.56 ($LAI=4.0$). Transmittance jumped from 0.38 to 0.52 ($LAI=1.5$). These increases represent the contributions of all higher scattering orders.

The residual imbalance of -5.4% and -3.0% is a known GL quadrature limitation: the Gauss-Legendre nodes, optimised for the flux integral $\int |\mu| \cdot I \, d\mu$, systematically underestimate the scalar irradiance integral $\int I \, d\mu$ for the multiply-scattered field at high ωL . The Reflected and Transmitted values are accurate. A more accurate absorbed estimate would require a moment-conserving quadrature.

Residual imbalance reflects GL quadrature limitation on scalar irradiance for high- ωL . Ref and Trans are accurate (boundary criterion also satisfied).

Assessment of the Successive Orders of Scattering Method

The SOS method has important strengths and real limitations that every practitioner should understand.

Strengths

- Simple and transparent: the algorithm maps directly onto physics. Each iteration adds one more scattering event, and the cumulative effect of multiple scattering is built up explicitly step by step.
- Intuitive: SOS is the best method for developing physical intuition about the scattering process. It makes explicit how the radiation field changes with each successive order and how the single-scattering albedo controls convergence speed.
- Exact in the limit: as $n \rightarrow \infty$ the SOS series converges to the exact solution of the RTE. No approximation is made in the physics — only in truncating the series at finite n .

Limitations

- How large should N (the number of scattering orders) be? There is no universal answer. It depends on both canopy depth (LAI) and the single-scattering albedo ω_L . For NIR canopies with large LAI, N can be as large as 100 in theory (in practice 8–16 suffice with a 1% threshold). One must iterate and check convergence, rather than choosing N a priori.
- The depth integrals $\int dL' Q(L', \Omega) P[\Omega, (L-L')]$ are computationally inconvenient in their analytical form. Evaluating them on a fine spatial grid for all (L, Ω) pairs is expensive, and the accuracy depends on the resolution chosen. A better approach — used in the companion code — replaces these continuous integrals with a discrete spatial grid (Method of Characteristics or Diamond-Difference sweep), which is both more efficient and more accurate for arbitrary source profiles.

To Do

- Download a detailed description of the SOS algorithm from the course website.
- Download the code `SOS_RT_canopy.py` and run options 2 (first-collision) and 3 (full SOS). Compare the energy balance tables for RED and NIR. Observe how many scattering orders are needed in each case and how the imbalance decreases as orders accumulate.